Small Angle Approximations

Exam code: 9MA0

Download PDF

Written by: Paul
Updated on 17 October 2025

SaveMyExams

Did this video help you? Yes No

Small angle approximations

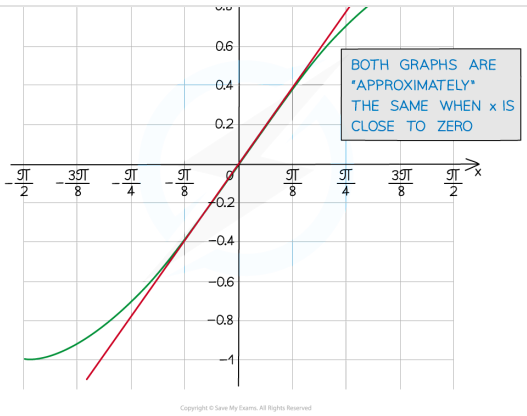
What is a small angle approximation?

- When an angle measured in **radians** is **very small**, you can **approximate** the value using **small angle approximations**
- These **only** apply when angles are measured in **radians**
- They can be applied to positive **and** negative small angles

What is the small angle approximation of sin θ?

$$\sin\theta \approx \theta$$

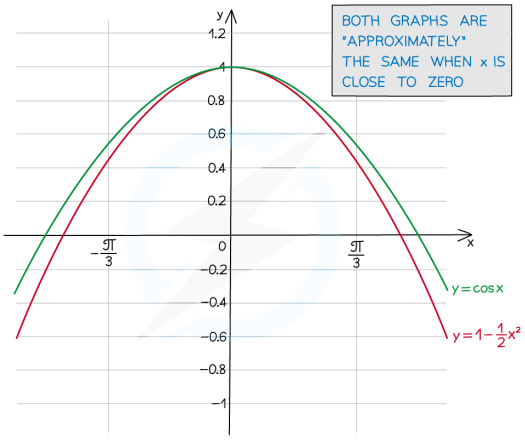
Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



Copyright © Save My Exams. All Rights Reserved

What is the small angle approximation of cos θ?

$$\cos\theta \approx 1 - \frac{1}{2}\theta^2$$

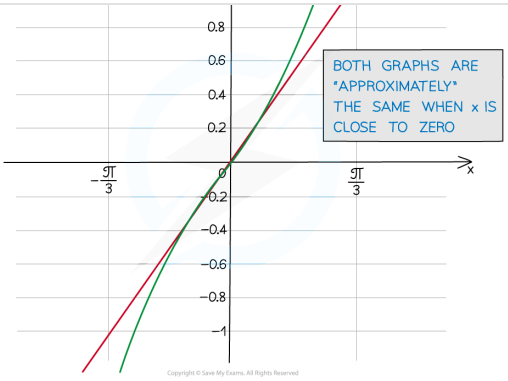


Copyright © Save My Exams. All Rights Reserved

- **y = cos θ** (near zero) is similar to a "negative quadratic" (parabola)

What is the small angle approximation of tan θ?

$$\tan\theta \approx \theta$$



How do I use small angle approximations?

- Replace sin θ, cos θ or tan θ with the appropriate approximation
- Given angles are often 2θ, 3θ, ...
  - Replace "θ" in the approximation by 2θ, 3θ, ...

e.g. GIVEN θ IS SMALL AND MEASURED IN RADIANS, FIND AN APPROXIMATION, IN TERMS OF θ FOR

a) $\cos 2\theta$

b) $\dfrac{\sin^2 3\theta + 2\theta}{\tan\theta}$

a) $\cos 2\theta \approx 1 - \frac{1}{2}(2\theta)^2$ ← "θ"=2θ

$= 1 - 2\theta^2$

$\boxed{\cos\theta \approx 1 - \frac{1}{2}\theta^2}$

"θ"=3θ

b) $\dfrac{\sin^2 3\theta + 2\theta}{\tan\theta} \approx \dfrac{(3\theta)^2 + 2\theta}{\theta}$

$\boxed{\sin\theta \approx \theta}$

$= \dfrac{9\theta^2 + 2\theta}{\theta}$

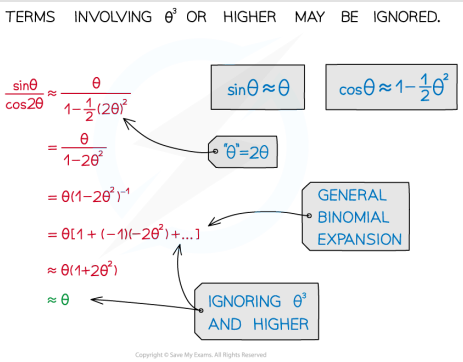
$\boxed{\tan\theta \approx \theta}$

$= \dfrac{\cancel{\theta}(9\theta + 2)}{\cancel{\theta}}$

$= 9\theta + 2$

Copyright © Save My Exams. All Rights Reserved

- **Binomial expansion** (see **GBE**) may be involved in more awkward expressions



Examiner Tips and Tricks

- Small angle approximations are given in the formula booklet.
- They can be used in proofs – particularly differentiation from first principles (see **First Principles Differentiation – Trigonometry**).

Worked Example

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder

Test yourself

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

Subjects

Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.
Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.